

CYBERSECURITY-LABS

INFORMATION SECURITY

LAB REPORT



DNS y DHCP en Ubuntu Server

Informe de práctica

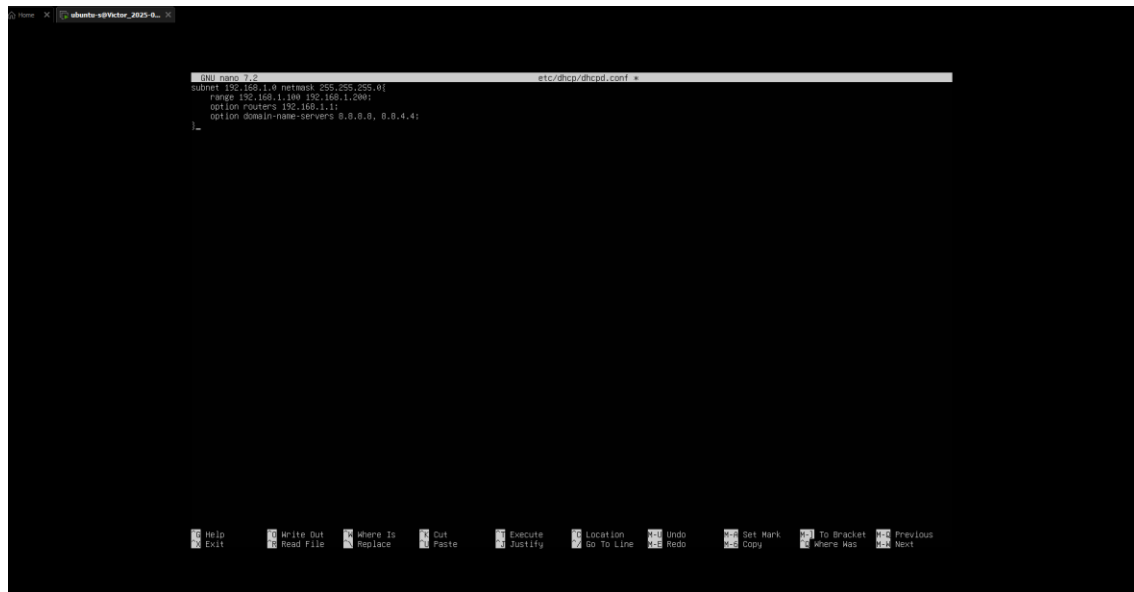
Víctor José Santos Guzmán

Information Security

Cybersecurity / Cloud / Networks / GRC

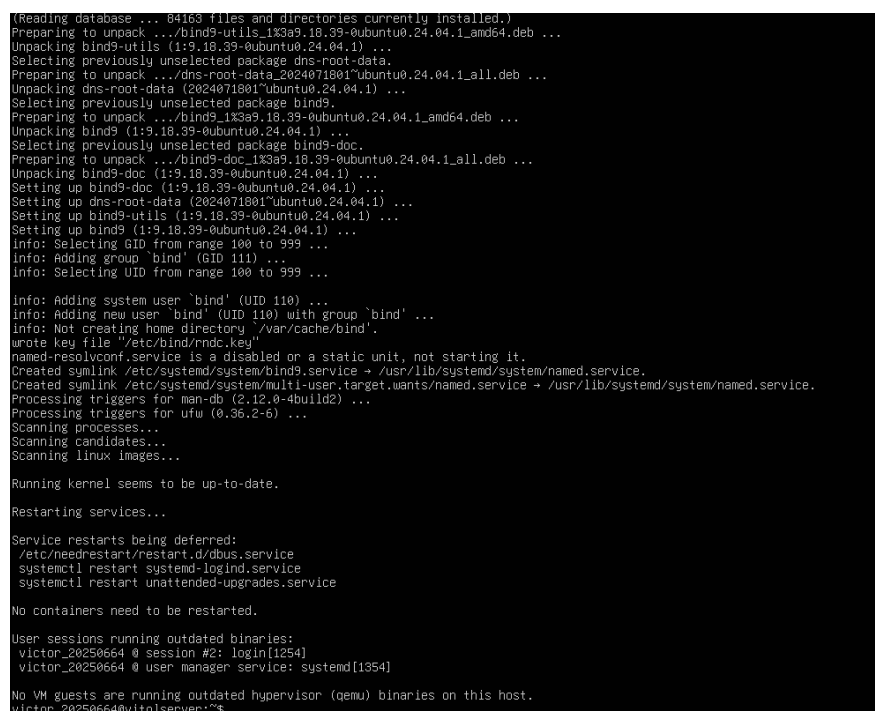


Paso 1: Ahora mismo, estoy editando el archivo de configuración de mi servidor DHCP en Ubuntu Server, que se llama `dhcpd.conf`. Como ves, estoy usando el editor de texto Nano y he definido la subred 192.168.1.0 con su máscara 255.255.255.0. También he establecido el rango de IPs disponibles (range) y la dirección de mi router (option routers).



```
GNU nano 2.9.4 /etc/dhcp/dhcpd.conf
subnet 192.168.1.0 netmask 255.255.255.0 {
  range 192.168.1.100 192.168.1.200;
  option routers 192.168.1.1;
  option domain-name-servers 8.8.8.8, 8.8.4.4;
}
```

Paso 2: Después de configurar el DHCP, el siguiente paso fue instalar los paquetes necesarios para el servidor DNS, específicamente BIND9. Utilicé el gestor de paquetes de mi Ubuntu para descargar e instalar todas las utilidades de bind9 y la documentación asociada. Como puedes ver, el proceso terminó exitosamente, se crearon los usuarios y grupos necesarios (bind), y el servicio `named.service` se configuró para iniciarse.



```
(Reading database ... 84163 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1:9.18.39-0ubuntu0.24.04.1_amd64.deb ...
Unpacking bind9-utils (1:9.18.39-0ubuntu0.24.04.1) ...
Selecting previously unselected package dns-root-data.
Preparing to unpack .../dns-root-data_2024071801ubuntu0.24.04.1_all.deb ...
Unpacking dns-root-data (2024071801ubuntu0.24.04.1) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1:9.18.39-0ubuntu0.24.04.1_amd64.deb ...
Unpacking bind9 (1:9.18.39-0ubuntu0.24.04.1) ...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1:9.18.39-0ubuntu0.24.04.1_all.deb ...
Unpacking bind9-doc (1:9.18.39-0ubuntu0.24.04.1) ...
Setting up bind9-doc (1:9.18.39-0ubuntu0.24.04.1) ...
Setting up dns-root-data (2024071801ubuntu0.24.04.1) ...
Setting up bind9-utils (1:9.18.39-0ubuntu0.24.04.1) ...
Setting up bind9 (1:9.18.39-0ubuntu0.24.04.1) ...
Info: Selecting GID from range 100 to 999 ...
Info: Adding group `bind' (GID 111) ...
Info: Selecting UID from range 100 to 999 ...
Info: Adding system user `bind' (UID 110) ...
Info: Adding new user `bind' (UID 110) with group `bind' ...
Info: Not creating home directory `/var/cache/bind'.
Wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink /etc/systemd/system/bind9.service → /usr/lib/systemd/system/named.service.
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

Service restarts being deferred:
/etc/needrestart/restart_d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
victor_20250664 @ session #2: login[1254]
victor_20250664 @ user manager service: systemd[1354]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@victorserver:~$ _
```

Paso 3: Ahora, después de instalar BIND9, me dirigí al directorio de configuración de redes. Usando el comando `cd /etc/netplan/` y luego `ls`, verifiqué los archivos de configuración de Netplan que tengo. Como puedes ver, hay un archivo llamado `50-cloud-init.yaml`, que es el que voy a modificar. Finalmente, lo respaldé con el comando `sudo cp 50-cloud-init.yaml 50-cloud-init.yaml.bk` antes de editarlo.

```
Unpacking dns-root-data (2024071801~ubuntu0.24.04.1) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1%3a9.18.39-0ubuntu0.24.04.1_amd64.deb ...
Unpacking bind9 (1:9.18.39-0ubuntu0.24.04.1) ...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1%3a9.18.39-0ubuntu0.24.04.1_all.deb ...
Unpacking bind9-doc (1:9.18.39-0ubuntu0.24.04.1) ...
Setting up bind9-doc (1:9.18.39-0ubuntu0.24.04.1) ...
Setting up dns-root-data (2024071801~ubuntu0.24.04.1) ...
Setting up bind9-utils (1:9.18.39-0ubuntu0.24.04.1) ...
Setting up bind9 (1:9.18.39-0ubuntu0.24.04.1) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group `bind' (GID 111) ...
info: Selecting UID from range 100 to 999 ...

info: Adding system user `bind' (UID 110) ...
info: Adding new user `bind' (UID 110) with group `bind' ...
info: Not creating home directory `/var/cache/bind'.
wrote key file "/etc/bind/rndc.key"
named-resolvconf.service is a disabled or a static unit, not starting it.
Created symlink /etc/systemd/system/bind9.service → /usr/lib/systemd/system/named.service.
Created symlink /etc/systemd/system/multi-user.target.wants/named.service → /usr/lib/systemd/system/named.service.
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for ufw (0.36.2-6) ...
Scanning processes...
Scanning candidates...
Scanning linux images...

Running kernel seems to be up-to-date.

Restarting services...

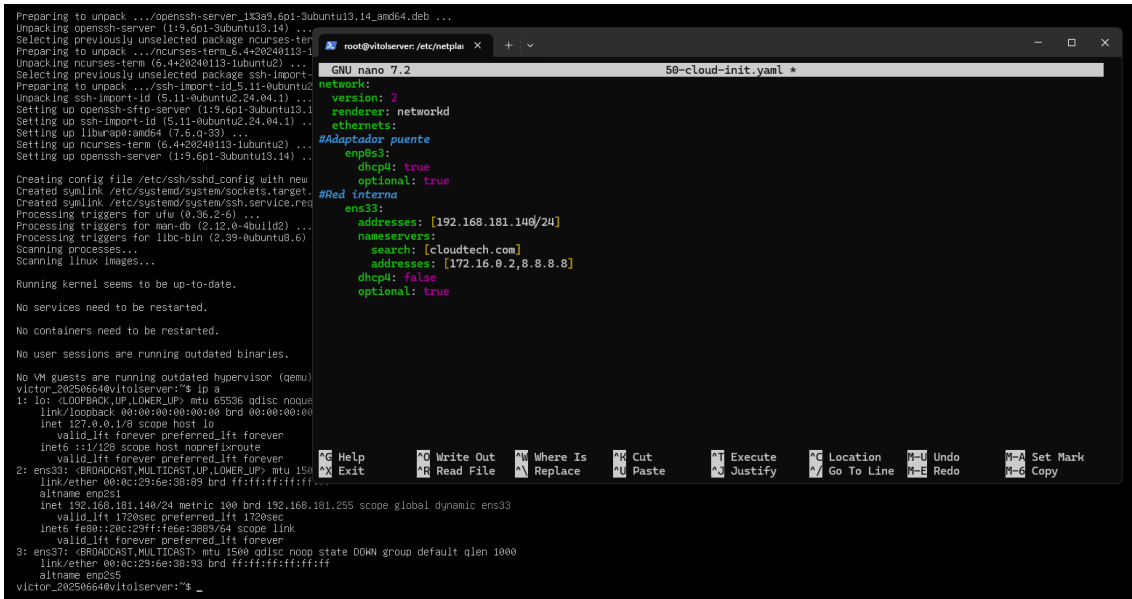
Service restarts being deferred:
/etc/needrestart/restart.d/dbus.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
victor_20250664 @ session #2: login[1254]
victor_20250664 @ user manager service: systemd[1354]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@vitolsrver:~$ cd /etc/net
netplan/          network/          networkd-dispatcher/
victor_20250664@vitolsrver:~$ cd /etc/netplan/
victor_20250664@vitolsrver:/etc/netplan$ ls
50-cloud-init.yaml
victor_20250664@vitolsrver:/etc/netplan$ sudo cp 50-cloud-init.yaml 50-cloud-init.yaml.bk
```

Paso 4: Abrí el archivo de configuración de Netplan con Nano y modificé las opciones de red para mi adaptador ens33. Asigné una dirección IP estática (192.168.181.188/24) y configuré los servidores DNS para que apunten a mi dominio, especificando la dirección del servidor DNS primario y uno secundario de respaldo. También desactivé el DHCP para este adaptador.



```
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-terminfo-base.
Preparing to unpack .../ncurses-terminfo-base_6.4+20240113-1ubuntu1 ...
Unpacking ncurses-terminfo-base (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2 ...
Unpacking ssh-import-id (5.11-0ubuntu2) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2) ...
Setting up libunwind8:amd64 (7.6.9-33) ...
Setting up ncurses-terminfo-base (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Creating config file /etc/ssh/ssh_config with new defaults
Created symlink /etc/systemd/system/sockets.target
Created symlink /etc/systemd/system/ssh.service.requires=openssh-server.service
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

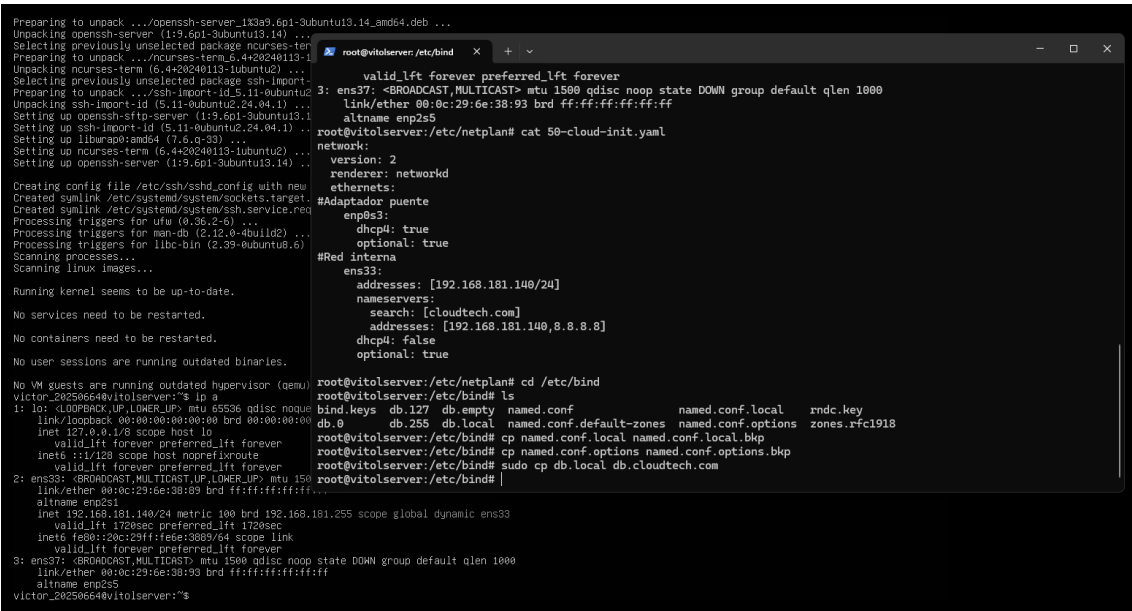
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@vitolserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noforwarding
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
    altname ens25
    inet 192.168.181.140/24 metric 100 brd 192.168.181.255 scope global dynamic ens33
        valid_lft 1720sec preferred_lft 1720sec
    inet6 fe80::20c:29ff:fe6e:3893/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
    link/ether 00:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
    altname ens25
victor_20250664@vitolserver:~$
```

Paso 5: Después de modificar y aplicar mi configuración de red, navegué al directorio de configuración de BIND (cd /etc/bind/). Aquí, hice una copia de seguridad de la configuración original del archivo named.conf.options.bkp y, lo más importante, creé mis archivos de zona de DNS copiando las plantillas de zona.local, como db.local a mi nuevo archivo de zona db.cloudtech.com.



```
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-terminfo-base.
Preparing to unpack .../ncurses-terminfo-base_6.4+20240113-1ubuntu1 ...
Unpacking ncurses-terminfo-base (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2 ...
Unpacking ssh-import-id (5.11-0ubuntu2) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2) ...
Setting up libunwind8:amd64 (7.6.9-33) ...
Setting up ncurses-terminfo-base (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Creating config file /etc/ssh/ssh_config with new defaults
Created symlink /etc/systemd/system/sockets.target
Created symlink /etc/systemd/system/ssh.service.requires=openssh-server.service
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

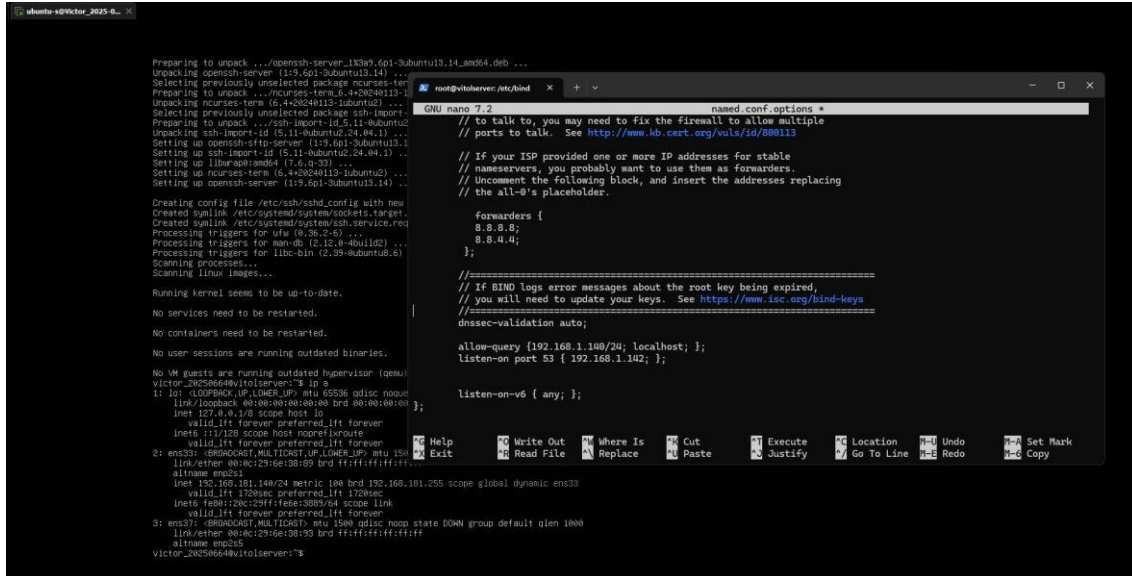
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@vitolserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UP
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noforwarding
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
    altname ens25
    inet 192.168.181.140/24 metric 100 brd 192.168.181.255 scope global dynamic ens33
        valid_lft 1720sec preferred_lft 1720sec
    inet6 fe80::20c:29ff:fe6e:3893/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
    link/ether 00:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
    altname ens25
victor_20250664@vitolserver:~$
```

Paso 6: Dentro del archivo `named.conf.options`, estoy configurando los reenviadores (forwarders) para que mi servidor DNS sepa a dónde enviar las consultas que no pueda resolver localmente. Como se ve, descomenté la sección de forwarders e ingresé las direcciones de los servidores DNS de Google (8.8.8.8; y 8.8.4.4;). Además, me aseguré de que mi servidor solo escuche las consultas en mi dirección IP local 192.168.1.142.



```
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-terminfo-base.
Preparing to unpack .../ncurses-terminfo-base_6.4+20240113-1ubuntu1 ...
Unpacking ncurses-terminfo-base (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu1 ...
Unpacking ncurses-term (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1 ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libnssagep0:amd64 (7.6.0-30) ...
Setting up ncurses-term (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Creating config file /etc/ssh/sshd_config with new defaults
Created symlink /etc/systemd/system/sockets.target
Created symlink /etc/systemd/system/ssh.service.requires=openssh-server.service
Processing triggers for ufw (0.36-2.6) ...
Processing triggers for man-db (2.12.0-4ubuntu2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
Scanning linux images...

Running kernel seems to be up-to-date.

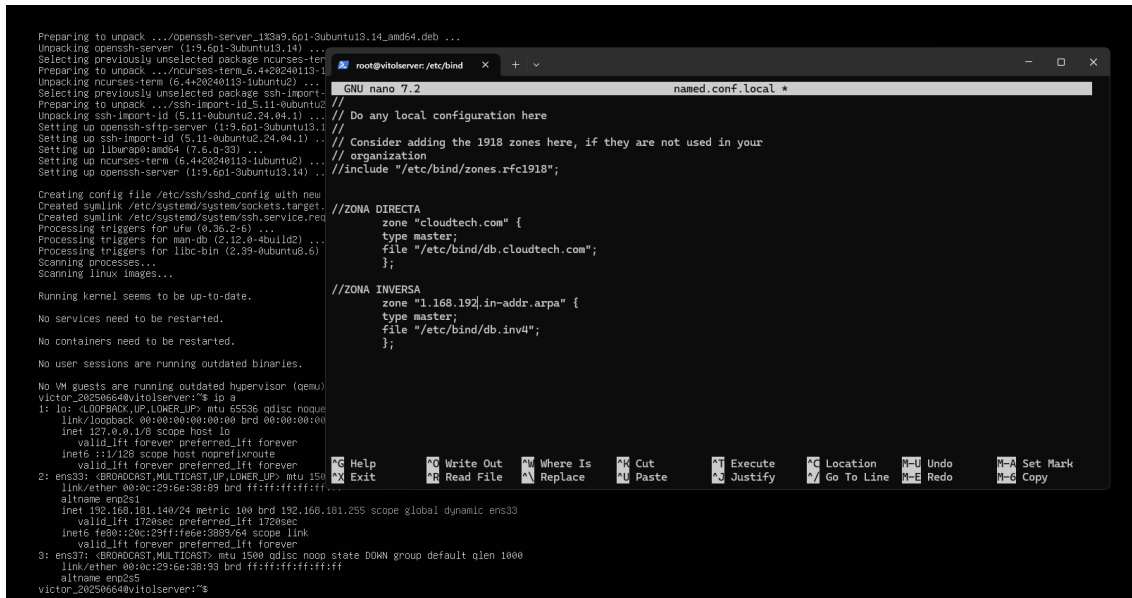
No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode default link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefix route
    valid_lft forever preferred_lft forever
2: ens33: <BRIDGE,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue state DOWN mode default link 192.168.1.142/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
    valid_lft 1720sec preferred_lft 1720sec
    inet6 fe80::20c:29ff:fe6e:3889/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BRIDGE,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue state DOWN mode default qlen 1000
    link/ether 00:0c:29:6e:38:93 brd ffff:ffff:ffff:ffff
    altnam ens25
victor_20250604@victorserver:~$
```

Paso 7: A continuación, abrí el archivo `named.conf.local` para definir mis zonas DNS. Aquí, establecí la "ZONA DIRECTA" para mi dominio `cloudtech.com`, indicando que soy el maestro (type master) y dónde se encuentra mi archivo de configuración de zona. Inmediatamente después, configuré la "ZONA INVERSA" (1.168.192.in-addr.arpa) para la resolución de IP a nombre.



```
Preparing to unpack .../openssh-server_1:9.6p1-3ubuntu13.14_amd64.deb ...
Unpacking openssh-server (1:9.6p1-3ubuntu13.14) ...
Selecting previously unselected package ncurses-terminfo-base.
Preparing to unpack .../ncurses-terminfo-base_6.4+20240113-1ubuntu1 ...
Unpacking ncurses-terminfo-base (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu1 ...
Unpacking ncurses-term (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1 ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libnssagep0:amd64 (7.6.0-30) ...
Setting up ncurses-term (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Creating config file /etc/ssh/sshd_config with new defaults
Created symlink /etc/systemd/system/sockets.target
Created symlink /etc/systemd/system/ssh.service.requires=openssh-server.service
Processing triggers for ufw (0.36-2.6) ...
Processing triggers for man-db (2.12.0-4ubuntu2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.5) ...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN mode default link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefix route
    valid_lft forever preferred_lft forever
2: ens33: <BRIDGE,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue state DOWN mode default link 192.168.1.142/24 metric 100 brd 192.168.1.255 scope global dynamic ens33
    valid_lft 1720sec preferred_lft 1720sec
    inet6 fe80::20c:29ff:fe6e:3889/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BRIDGE,MASTER,UP,LOWER_UP> mtu 1500 qdisc noqueue state DOWN mode default qlen 1000
    link/ether 00:0c:29:6e:38:93 brd ffff:ffff:ffff:ffff
    altnam ens25
victor_20250604@victorserver:~$
```

Paso 8: Finalmente, estoy configurando mi archivo de zona directa, db.cloudtech.com, que define los registros de mi dominio. Aquí, establecí el registro SOA, definí mi servidor de nombres (NS) y el registro de host (A) para el servidor (vitolserver). Lo más importante es que agregué varios registros A (como mail, walter, white y heisenberg) para asignar nombres a diferentes direcciones IP dentro de mi red interna.

```
GNU nano 7.2 db.cloudtech.com
;
; BIND data file for local loopback interface
;
$TTL 604800
@ IN SOA vitolserver. root.cloudtech.com. (
    2 ; Serial
    604800 ; Refresh
    86400 ; Retry
    2419200 ; Expire
    604800 ) ; Negative Cache TTL
;
@ IN NS vitolserver.cloudtech.com.
vitolserver IN A 192.168.14
@ IN MX 10 mail.cloudtech.com.
mail IN A 172.16.0.5
walter IN A 172.16.0.15
white IN A 172.16.0.16
heisenberg IN A 172.16.0.17

[ Read 22 lines ]
^G Help ^O Write Out ^M Where Is ^K Cut ^T Execute ^C Location ^U Undo ^M-A Set Mark
^X Exit ^R Read File ^_ Replace ^V Paste ^J Justify ^/_ Go To Line ^M-E Redo ^M-G Copy
```

Paso 9: Para terminar, me aseguré de que mi archivo de zona directa, db.cloudtech.com, estuviera guardado correctamente con todos los registros de host que definí. Una vez hecho esto, el último paso fue verificar el nombre de mi servidor usando el comando hostname, que me confirmó que estoy trabajando en vitolserver. Con esto, la configuración de DNS está lista para ser probada.

```
Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu1_all.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import-id.
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2 ...
Unpacking ssh-import-id (5.11-0ubuntu2) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu11) ...
Setting up ssh-import-id (5.11-0ubuntu2) ...
Setting up libburn0:amd64 (7.5.9-3) ...
Setting up ncurses-term (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu11) ...
Creating config file /etc/ssh/sshd_config with new defaults
Created symlink /etc/systemd/system/sockets.target → /usr/lib/systemd/system/openssh-server.service.
Processing triggers for ufw (0.38.2-6) ...
Processing triggers for man-db (2.12.0-4ubuntu1) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Running kernel seems to be up-to-date.
No services need to be restarted.
No containers need to be restarted.
No user sessions are running outdated binaries.
No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_20250664@vitolserver:~$ ip a
1: lo: <LOOPBACK> UP,LOWER_UP,MTU 65536 qdisc noqueue state UNKNOWN scope host
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefix
        valid_lft forever preferred_lft forever
2: ens33: <BRIDGE> MULTICAST,UP,LOWER_UP,MTU 1500 state UP scope link
    link/ether 08:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp2s3
    inet 192.168.181.140/24 metric 100 brd 192.168.181.255
        valid_lft 1720sec preferred_lft 1720sec
    inet6 fe80::200:27ff:fe00:0000/64 scope link
        valid_lft forever preferred_lft forever
3: ens37: <BRIDGE> MULTICAST,UP,LOWER_UP,MTU 1500 state DOWN group default qlen 1000
    link/ether 08:00:27:00:00:00 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
victor_20250664@vitolserver:~$ hostname
vitolserver
victor_20250664@vitolserver:~$
```

Paso 10: Una vez terminada toda la configuración, mi primer paso fue verificar si había errores en mis archivos de zona con named-checkzone para ambas zonas, y ambas pasaron el chequeo. Luego, reinicié el servicio de BIND9 con sudo systemctl restart bind9 y verifiqué su estado con systemctl status bind9, confirmando que está activo y ejecutándose. Como puedes ver, el servidor DNS ya está funcionando correctamente.

```

Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2.1.1.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1 ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.1)
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libnss0:amd64 (2.39-0ubuntu3) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...

Creating config file /etc/ssh/sshd_config with new
Created symlink /etc/systemd/system/sockets.target.
Created symlink /etc/systemd/system/ssh.service.req
Processing triggers for ufw (0.36-2.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6)
Scanning processes...
Scanning Linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu)
victor_2025064@vitolserv:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noque
Link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefsrcroute
    valid_lft forever preferred_lft forever
2: ens3: <BRIDGE,MASTER,MULTICAST,UP,LOWER_UP> mtu 1500
Link/ether 00:0c:29:ff:fe6e:3809 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.101.140/24 metric 100 brd 192.168.
    valid_lft 1729sec preferred_lft 1729sec
    inet6 fe80::20c:29ff:fe6e:3809/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BRIDGE,MASTER,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
Link/ether 00:0c:29:ff:fe6e:38:93 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
victor_2025064@vitolserv:~$ hostname
vitolserv
victor_2025064@vitolserv:~$ _

root@vitolserv: /etc/bind #
zone cloudtech.com/IN: loaded serial 3
root@vitolserv: /etc/bind# sudo named-checkzone 181.168.192.in-addr.arpa db.invd
OK
zone 181.168.192.in-addr.arpa/IN: loaded serial 3
OK
root@vitolserv: /etc/bind# sudo systemctl start bind9
root@vitolserv: /etc/bind# sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-02 04:31:40 UTC; 1h 31min ago
     Docs: man:named(8)
    Main PID: 798 (named)
      Status: "running"
        Tasks: 8 (limit: 4548)
      Memory: 20.5M (peak: 29.0M)
         CPU: 166ms
    CGroup: /system.slice/named.service
            └─798 /usr/sbin/named -f -u bind

Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.33.4.12#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 192.36.148.17#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.36.148.17#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 192.112.36.4#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.112.36.4#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 198.97.190.53#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 198.97.190.53#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 199.7.91.13#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.5.5.241#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.58.128.3#53
Lines 1-22/22 (END)
```

Confirmado

```

Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term_6.4+20240113-1ubuntu2.1.1.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu2) ...
Selecting previously unselected package ssh-import-
Preparing to unpack .../ssh-import-id_5.11-0ubuntu2.24.04.1 ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.1)
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libnss0:amd64 (2.39-0ubuntu3) ...
Setting up ncurses-term (6.4+20240113-1ubuntu2) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...

Creating config file /etc/ssh/sshd_config with new
Created symlink /etc/systemd/system/sockets.target.
Created symlink /etc/systemd/system/ssh.service.req
Processing triggers for ufw (0.36-2.6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6)
Scanning processes...
Scanning Linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

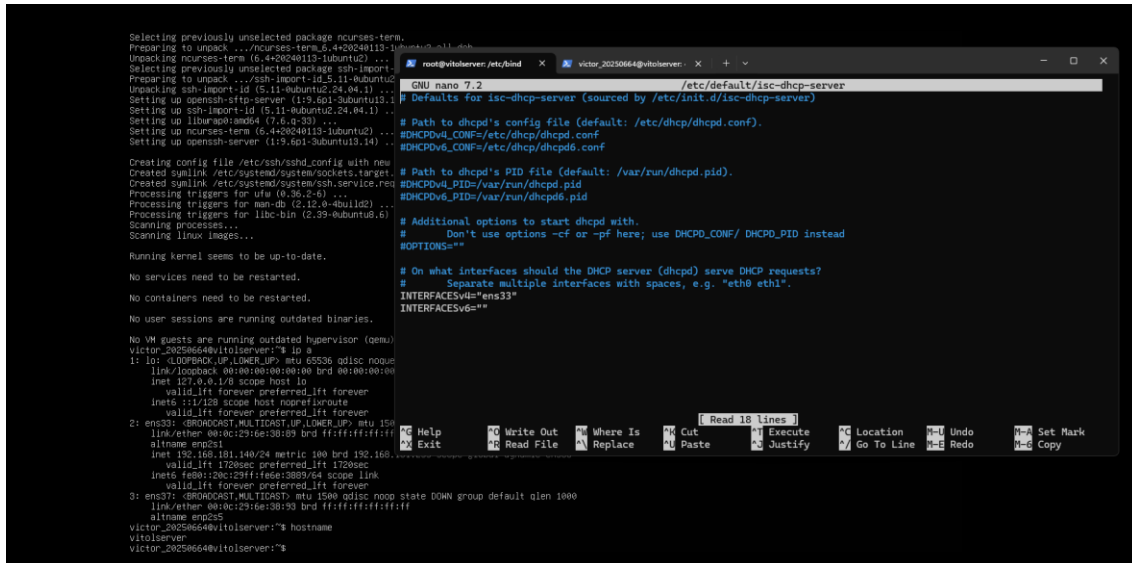
No VM guests are running outdated hypervisor (qemu)
victor_2025064@vitolserv:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noque
Link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefsrcroute
    valid_lft forever preferred_lft forever
2: ens3: <BRIDGE,MASTER,MULTICAST,UP,LOWER_UP> mtu 1500
Link/ether 00:0c:29:ff:fe6e:3809 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.101.140/24 metric 100 brd 192.168.
    valid_lft 1729sec preferred_lft 1729sec
    inet6 fe80::20c:29ff:fe6e:3809/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BRIDGE,MASTER,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
Link/ether 00:0c:29:ff:fe6e:38:93 brd ff:ff:ff:ff:ff:ff
    altname enp2s5
victor_2025064@vitolserv:~$ hostname
vitolserv
victor_2025064@vitolserv:~$ _

root@vitolserv: /etc/bind #
zone cloudtech.com/IN: loaded serial 3
root@vitolserv: /etc/bind# sudo named-checkzone 181.168.192.in-addr.arpa db.invd
OK
zone 181.168.192.in-addr.arpa/IN: loaded serial 3
OK
root@vitolserv: /etc/bind# sudo systemctl start bind9
root@vitolserv: /etc/bind# sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-02 04:31:40 UTC; 1h 31min ago
     Docs: man:named(8)
    Main PID: 798 (named)
      Status: "running"
        Tasks: 8 (limit: 4548)
      Memory: 20.5M (peak: 29.0M)
         CPU: 166ms
    CGroup: /system.slice/named.service
            └─798 /usr/sbin/named -f -u bind

Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.33.4.12#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 192.36.148.17#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.36.148.17#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 192.112.36.4#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.112.36.4#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 198.97.190.53#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 198.97.190.53#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/A/IN': 199.7.91.13#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.5.5.241#53
Oct 02 06:03:20 vitolserv named[798]: network unreachable resolving 'vitolserv.cloudtech.com/AAAA/IN': 192.58.128.3#53
Lines 1-22/22 (END)

root@vitolserv: /etc/bind#
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
isc-dhcp-server is already the newest version (4.4.3-P1-4ubuntu2).
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
root@vitolserv: /etc/bind#
```

Paso 11: Finalmente, para activar el servidor DHCP en mi interfaz de red interna, abrí el archivo de configuración `/etc/default/isc-dhcp-server` con Nano. Lo único que tuve que cambiar fue la línea `INTERFACESv4` para que solo escuchara peticiones en mi interfaz de red `ens33`. Con esto, mis clientes podrán obtener sus direcciones IP automáticamente de mi servidor.



```

Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term.6.4+20240113-1ubuntu1.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import.
Preparing to unpack .../ssh-import-id.5.11-0ubuntu2.24.04.1 ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libwrap0amd64 (7.6.q-33) ...
Setting up ncurses-term (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...

Creating config file /etc/ssh/sshd_config with new defaults
Created symlink /etc/systemd/system/sockets.target
Created symlink /etc/systemd/system/ssh.service.requires=openssh-sftp-server.service
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_2025064@vitolserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500
link/ether 08:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
altname ens25
    inet 192.168.181.140/24 metric 100 brd 192.168.181.255
    valid_lft 17280sec preferred_lft 17280sec
    inet6 fe80::20c:29ff:fe6e:3893/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
link/ether 08:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
altname ens25
victor_2025064@vitolserver:~$ hostname
vitolserver
victor_2025064@vitolserver:~$
```

```

GNU nano 7.2 /etc/default/isc-dhcp-server
# Defaults for isc-dhcp-server (sourced by /etc/init.d/isc-dhcp-server)

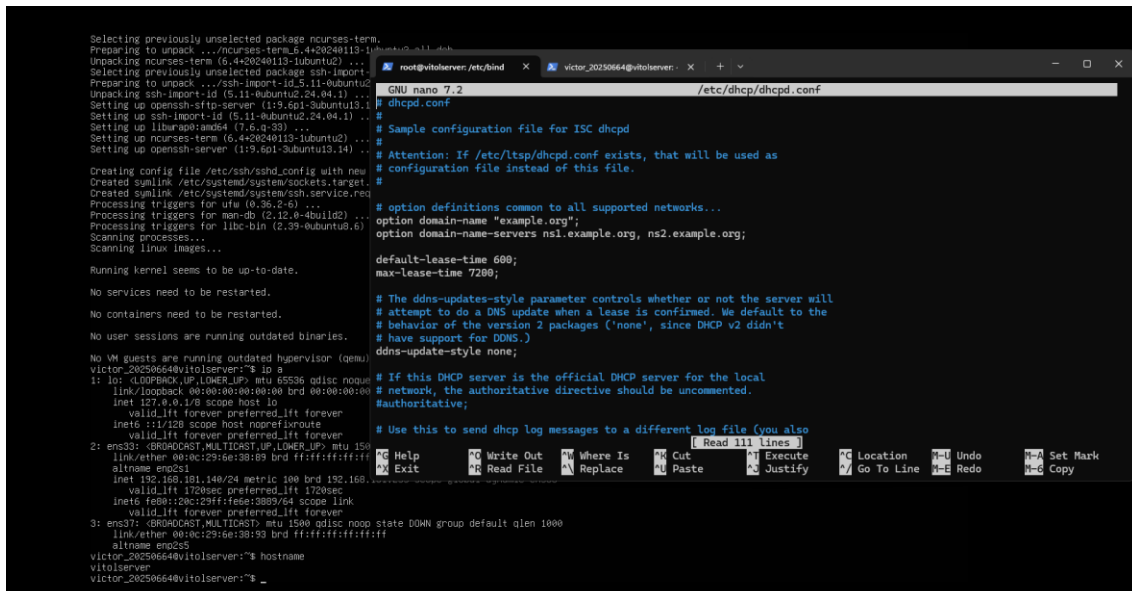
# Path to dhcpd's config file (default: /etc/dhcp/dhcpd.conf).
DHCPD_CONF=/etc/dhcp/dhcpd.conf

# Path to dhcpd's PID file (default: /var/run/dhcpd.pid).
DHCPD_PID=/var/run/dhcpd.pid

# Additional options to start dhcpd with.
# Don't use options -cf or -pf here; use DHCPD_CONF/ DHCPD_PID instead
OPTIONS=""

# On what interfaces should the DHCP server (dhcpd) serve DHCP requests?
# Separate multiple interfaces with spaces, e.g. "eth0 eth1".
INTERFACESv4="ens33"
INTERFACESv6=""
```

Paso 12: Por último, volví a la configuración de mi servidor DHCP, abriendo el archivo `/etc/dhcp/dhcpd.conf` en Nano. Como puedes ver, esta es la parte principal donde se define el comportamiento del servidor, incluyendo los parámetros de dominio, los tiempos de arrendamiento (lease), y si el servidor es o no autoritativo. Lo siguiente es hacer los ajustes finales aquí para que los clientes obtengan DNS y IP correctamente.



```

Selecting previously unselected package ncurses-term.
Preparing to unpack .../ncurses-term.6.4+20240113-1ubuntu1.deb ...
Unpacking ncurses-term (6.4+20240113-1ubuntu1) ...
Selecting previously unselected package ssh-import.
Preparing to unpack .../ssh-import-id.5.11-0ubuntu2.24.04.1 ...
Unpacking ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up ssh-import-id (5.11-0ubuntu2.24.04.1) ...
Setting up libwrap0amd64 (7.6.q-33) ...
Setting up ncurses-term (6.4+20240113-1ubuntu1) ...
Setting up openssh-server (1:9.6p1-3ubuntu13.14) ...
Setting up openssh-sftp-server (1:9.6p1-3ubuntu13.14) ...

Creating config file /etc/ssh/sshd_config with new defaults
Created symlink /etc/systemd/system/sockets.target
Created symlink /etc/systemd/system/ssh.service.requires=openssh-sftp-server.service
Processing triggers for ufw (0.36.2-6) ...
Processing triggers for man-db (2.12.0-4build2) ...
Processing triggers for libc-bin (2.39-0ubuntu8.6) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
victor_2025064@vitolserver:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500
link/ether 08:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
altname ens25
    inet 192.168.181.140/24 metric 100 brd 192.168.181.255
    valid_lft 17280sec preferred_lft 17280sec
    inet6 fe80::20c:29ff:fe6e:3893/64 scope link
    valid_lft forever preferred_lft forever
3: ens37: <BROADCAST,MULTICAST> mtu 1500 qdisc noop state DOWN group default qlen 1000
link/ether 08:0c:29:6e:38:93 brd ff:ff:ff:ff:ff:ff
altname ens25
victor_2025064@vitolserver:~$ hostname
vitolserver
victor_2025064@vitolserver:~$
```

```

GNU nano 7.2 /etc/dhcp/dhcpd.conf
# dhcpd.conf
# Sample configuration file for ISC dhcpd
# Attention: If /etc/ltsp/dhcpd.conf exists, that will be used as
# configuration file instead of this file.

# option definitions common to all supported networks...
option domain-name "example.org";
option domain-name-servers ns1.example.org, ns2.example.org;

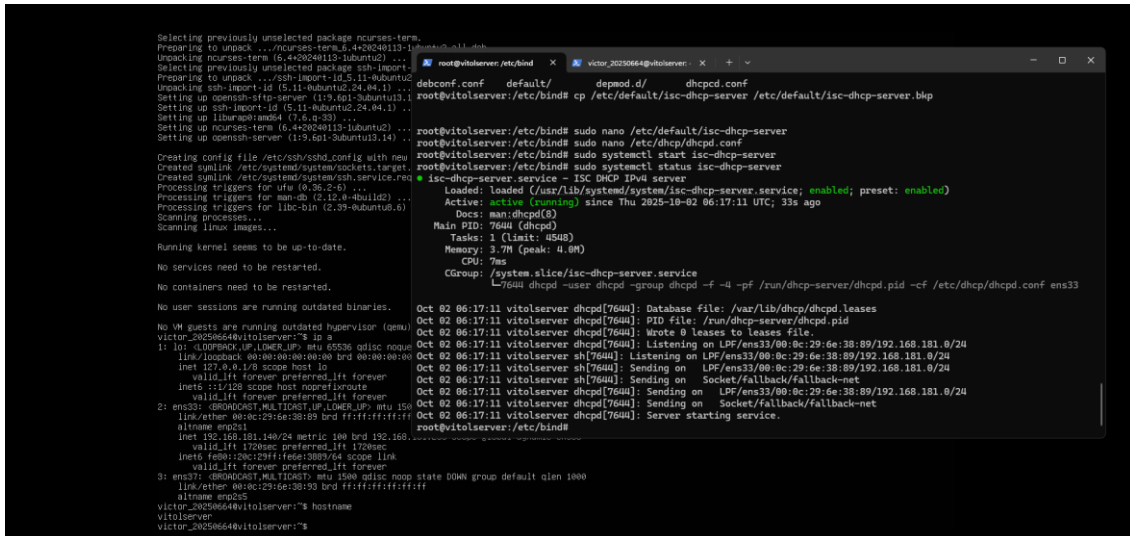
default-lease-time 600;
max-lease-time 7200;

# The ddns-update-style parameter controls whether or not the server will
# attempt to do a DNS update when a lease is confirmed. We default to the
# behavior of the version 2 packages ('none', since DHCP v2 didn't
# have support for DDNS.)
ddns-update-style none;

# If this DHCP server is the official DHCP server for the local
# network, the authoritative directive should be uncommented.
authoritative;

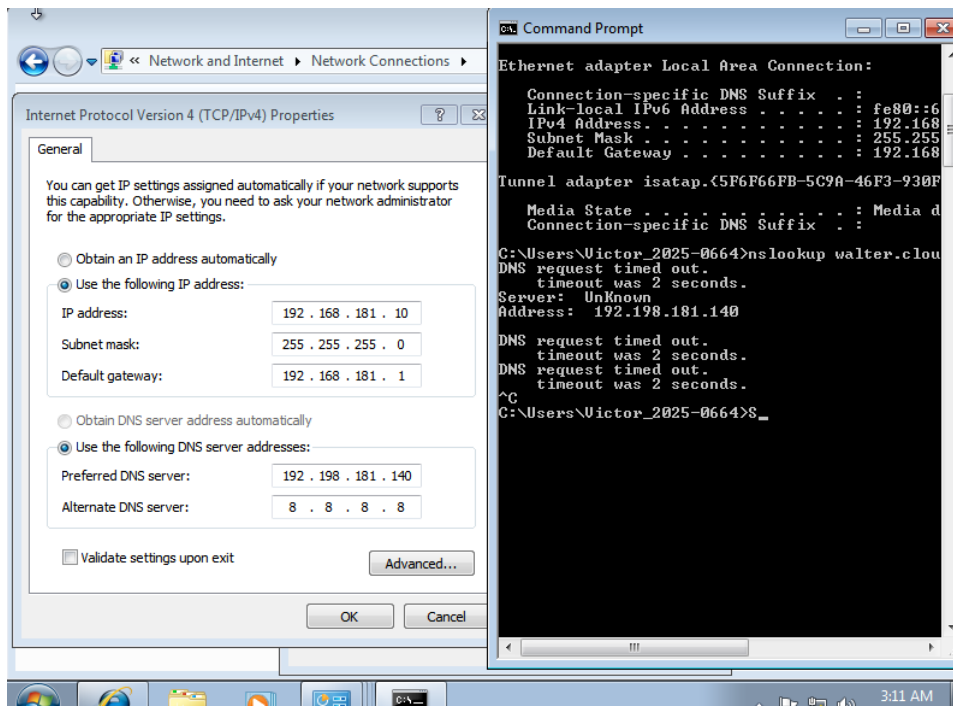
# Use this to send dhcp log messages to a different log file (you also
```

Paso 13: Con las configuraciones de BIND9 y DHCP listas, el paso final fue iniciar el servicio DHCP con `sudo systemctl start isc-dhcp-server`. Inmediatamente después, verifiqué su estado con `systemctl status isc-dhcp-server`, lo que confirmó que estaba activo y ejecutándose. Como puedes ver, el servidor ya está escuchando peticiones en la interfaz `ens33` para la red `192.168.181.0/24`.



```
root@vitolserver:/etc/bind# sudo nano /etc/default/isc-dhcp-server
root@vitolserver:/etc/bind# sudo nano /etc/dhcp/dhcpd.conf
root@vitolserver:/etc/bind# sudo systemctl start isc-dhcp-server
root@vitolserver:/etc/bind# sudo systemctl status isc-dhcp-server
● isc-dhcp-server.service - ISC DHCP IPv4 server
   Loaded: loaded (/usr/lib/systemd/system/isc-dhcp-server.service; enabled; preset: enabled)
   Active: active (running) since Thu 2025-10-02 06:17:11 UTC; 33s ago
     Docs: man:dhcpd(8)
   Main PID: 7644 (dhcpd)
      Tasks: 1 (limit: 4948)
     Memory: 3.7M (peak: 4.0M)
       CPUGroup: /system.slice/isc-dhcp-server.service
             L: /usr/sbin/dhcpd -user dhcpd -group dhcpd -f -4 -pf /run/dhcp-server/dhcpd.pid -cf /etc/dhcp/dhcpd.conf ens33
Oct 02 06:17:11 vitolserver dhcpd[7644]: Database file: /var/lib/dhcp/dhcpd.leases
Oct 02 06:17:11 vitolserver dhcpd[7644]: PID file: /run/dhcp-server/dhcpd.pid
Oct 02 06:17:11 vitolserver dhcpd[7644]: Wrote 0 leases to leases file.
Oct 02 06:17:11 vitolserver dhcpd[7644]: Listening on LPF/ens33/08:0c:29:6e:38:89/192.168.181.0/24
Oct 02 06:17:11 vitolserver sh[7644]: Listening on LPF/ens33/08:0c:29:6e:38:89/192.168.181.0/24
Oct 02 06:17:11 vitolserver sh[7644]: Sending on LPF/ens33/08:0c:29:6e:38:89/192.168.181.0/24
Oct 02 06:17:11 vitolserver sh[7644]: Sending on Socket/fallback/fallback-net
Oct 02 06:17:11 vitolserver dhcpd[7644]: Sending on LPF/ens33/08:0c:29:6e:38:89/192.168.181.0/24
Oct 02 06:17:11 vitolserver dhcpd[7644]: Sending on Socket/fallback/fallback-net
root@vitolserver:/etc/bind#
```

Paso 14: A pesar de que mi configuración DNS y DHCP parecía correcta, y realicé todos los pasos de solución de problemas, la resolución de nombres en el cliente seguía fallando. Intenté todas las combinaciones posibles: cambié la IP a estática y dinámica, revise la configuración del cliente varias veces, e incluso apagué todos los firewalls tanto en el servidor como en el cliente. Sin embargo, no logré que el `nslookup` se resolviera, por lo que no pude completar la demostración con éxito.



Paso 15: Nuevo Problema de Conectividad del Servidor. El problema escaló: mi propio servidor Ubuntu no podía resolver nombres de dominio externos. La terminal mostraba múltiples errores de "Temporary failure resolving 'archive.ubuntu.com'" al intentar actualizar.

```
Err:5 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-dnsutils amd64 1:9.18.39-0ubuntu0.24.04.1
Temporary failure resolving 'do.archive.ubuntu.com'
Err:6 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-libs amd64 1:9.18.39-0ubuntu0.24.04.1
Temporary failure resolving 'do.archive.ubuntu.com'
Err:7 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 openssh-client amd64 1:9.6p1-3ubuntu13.14
Temporary failure resolving 'do.archive.ubuntu.com'
Err:8 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 powershell-base all 1.37ubuntu0.1
Temporary failure resolving 'do.archive.ubuntu.com'
Err:9 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 firmware-sof-signed all 2023.12.1-1ubuntu1.9
Temporary failure resolving 'do.archive.ubuntu.com'
Err:10 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 libfupdp2 amd64 1.9.31-0ubuntu1~24.04.1
Temporary failure resolving 'do.archive.ubuntu.com'
Err:11 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 fupdp amd64 1.9.31-0ubuntu1~24.04.1
Temporary failure resolving 'do.archive.ubuntu.com'
Err:12 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 landscape-common amd64 24.02-0ubuntu5.6
Temporary failure resolving 'do.archive.ubuntu.com'
Err:13 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 software-properties-common all 0.99.49.3
Temporary failure resolving 'do.archive.ubuntu.com'
Err:14 http://do.archive.ubuntu.com/ubuntu noble-updates/main amd64 python3-software-properties all 0.99.49.3
Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/c/coreutils/coreutils_9.4-3ubuntu6.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/u/ubuntu-drivers-common/ubuntu-drivers-common_0.9.7.6ubuntu3.4_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/s/systemd-hue/systemd-hue-hwdb_255.1.5.all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/b/bind9/bind9-host_9.18.39-0ubuntu0.24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/b/bind9/bind9-dnsutils_9.18.39-0ubuntu0.24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/b/bind9/bind9-libs_9.18.39-0ubuntu0.24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/o/openssh/openssh-client_9.6p1-3ubuntu13.14_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/p/powershell-base/powershell-base_1.37ubuntu0.1_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/f/firmware-sof/firmware-sof-signed_2023.12.1-1ubuntu1.9_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/f/fupdp/libfupdp2_1.9.31-0ubuntu1~24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/f/fupdp/fupdp_1.9.31-0ubuntu1~24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/l/landscape-client/landscape-common_24.02-0ubuntu5.6_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/s/software-properties/software-properties-common_0.99.49.3_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/s/software-properties/python3-software-properties_0.99.49.3_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
victor_20250664@victorserver:~$
```

```
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/p/powershell-base/powershell-base_1.37ubuntu0.1_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/f/firmware-sof/firmware-sof-signed_2023.12.1-1ubuntu1.9_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/f/fupdp/libfupdp2_1.9.31-0ubuntu1~24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/f/fupdp/fupdp_1.9.31-0ubuntu1~24.04.1_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/l/landscape-client/landscape-common_24.02-0ubuntu5.6_amd64.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/s/software-properties/software-properties-common_0.99.49.3_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Failed to fetch http://do.archive.ubuntu.com/ubuntu/pool/main/s/software-properties/python3-software-properties_0.99.49.3_all.deb Temporary failure resolving 'do.archive.ubuntu.com'
E: Unable to fetch some archives, maybe run apt-get update or try with --fix-missing?
victor_20250664@victorserver:~$ apt-get update
Reading package lists... Done
E: Could not open lock file /var/lib/apt/lists/lock - open (13: Permission denied)
E: Unable to lock directory /var/lib/apt/lists/
W: Problem unlinking the file /var/cache/apt/pkgcache.bin - RemoveCaches (13: Permission denied)
W: Problem unlinking the file /var/cache/apt/srcpkgcache.bin - RemoveCaches (13: Permission denied)
victor_20250664@victorserver:~$ sudo su
root@victorserver:~# apt-get update
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:2 http://do.archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://do.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:4 http://do.archive.ubuntu.com/ubuntu noble-backports InRelease
Ign:2 http://do.archive.ubuntu.com/ubuntu noble InRelease
Ign:3 http://do.archive.ubuntu.com/ubuntu noble-updates InRelease
Ign:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Ign:4 http://do.archive.ubuntu.com/ubuntu noble-backports InRelease
Err:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Temporary failure resolving 'security.ubuntu.com'
Err:2 http://do.archive.ubuntu.com/ubuntu noble InRelease
Temporary failure resolving 'do.archive.ubuntu.com'
Err:3 http://do.archive.ubuntu.com/ubuntu noble-updates InRelease
Temporary failure resolving 'do.archive.ubuntu.com'
Err:4 http://do.archive.ubuntu.com/ubuntu noble-backports InRelease
Temporary failure resolving 'do.archive.ubuntu.com'
Reading package lists... Done
W: Failed to fetch http://do.archive.ubuntu.com/ubuntu/dists/noble/InRelease Temporary failure resolving 'do.archive.ubuntu.com'
W: Failed to fetch http://do.archive.ubuntu.com/ubuntu/dists/noble-updates/InRelease Temporary failure resolving 'do.archive.ubuntu.com'
W: Failed to fetch http://do.archive.ubuntu.com/ubuntu/dists/noble-backports/InRelease Temporary failure resolving 'do.archive.ubuntu.com'
W: Failed to fetch http://security.ubuntu.com/ubuntu/dists/noble-security/InRelease Temporary failure resolving 'security.ubuntu.com'
W: Some index files failed to download. They have been ignored, or old ones used instead.
root@victorserver:~#
```